Main Results and a Prediction Extension

Replicating DellaVigna and Pope (2018)

Qianyu Chen | Replication note

1. Replication of the main results

I replicate Table 3 and both panels of Table 5 in Python, using the authors' released cleaned data: 9,861 MTurk workers and 208 experts. The data-cleaning audit is reported separately. For Table 5, my estimates and the published ones appear in adjacent columns.

1.1 Table 3: effort and expert forecasts

Table 3 (Replication): Findings by treatment: effort and expert forecasts

Category	Treatment description	N	Mean effort (s.e.)	Mean forecast	Std. dev. forecast	Actual − forecast (s.e.)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Piece rate	No piece rate	540	1521 (31.23)		Benchmark	
	1 cent per 100 points	558	2029 (27.47)		Benchmark	
	10 cents per 100 points	566	2175 (24.28)		Benchmark	
	4 cents per 100 points	562	2132 (26.42)	2057	120.87	75 (27.71)
Pay enough or don't pay	1 cent per 1,000 points	538	1883 (28.61)	1657	262.00	226 (33.89)
Social preferences: charity	Charity receives 1 cent per 100 points	554	1907 (26.85)	1894	202.20	13 (30.29)
	Charity receives 10 cents per 100 points	549	1918 (25.93)	1997	196.75	−78 (29.30)
Social preferences: gift exchange	Unconditional gift of 40 cents	545	1602 (29.77)	1709	207.12	−108 (33.05)
Discounting	1 cent per 100 points, paid after 2 weeks	544	2004 (27.38)	1933	142.03	72 (29.10)
	1 cent per 100 points, paid after 4 weeks	550	1970 (28.68)	1895	162.54	75 (30.81)
Gains versus losses	Gain 40 cents for reaching 2,000 points	545	2136 (24.66)	1955	149.90	181 (26.76)
	Lose 40 cents unless reaching 2,000 points	532	2155 (23.09)	2002	143.57	152 (25.14)
	Gain 80 cents for reaching 2,000 points	532	2188 (22.99)	2007	131.93	181 (24.74)
Risk aversion and probability weighting	1% chance of $1 per 100 points	555	1896 (28.44)	1967	253.43	−71 (33.43)
	50% chance of 2 cents per 100 points	568	1977 (24.73)	1941	179.27	36 (27.68)
Social comparisons	Information that others scored above 2,000	526	1848 (32.14)	1877	209.48	−29 (35.27)
Ranking	Feedback on relative performance	543	1761 (30.63)	1850	234.28	−89 (34.67)
Task significance	Request to try as hard as possible	554	1740 (28.76)	1757	230.15	−17 (32.89)

Notes: Categories, treatment order, and numerical columns follow published Table 3; treatment descriptions are shortened. Parentheses contain standard errors. Forecast standard deviations measure dispersion across 208 experts. The first three treatments were supplied to experts as benchmarks. Source: my Python output using the released cleaned data.

Treatment counts, mean effort and its standard error, and the mean and dispersion of expert forecasts all match the paper at its reported precision. The only differences are in column (7). Four actual-minus-forecast gaps differ by one point: charity at 10 cents (−78 versus −79), gift exchange (−108 versus −107), the two-week delay (72 versus 71) and the 40-cent loss (152 versus 153). All four come from rounding: I subtract unrounded means, and subtracting the rounded means reproduces the published values. The standard error of the 4-cent gap is 27.71, versus 27.72 in the paper.

The descriptive findings are therefore the same as in the paper. Mean effort rises from 1,521 without a piece rate to 2,029 at 1 cent per 100 points and 2,175 at 10 cents. Experts underpredict effort when pay is very low (by 226

points at 1 cent per 1,000 points) and in the gain and loss treatments. They overpredict the response to a larger charitable return.

1.2 Table 5A: benchmark parameters and predictions

I estimate power and exponential costs using the 0-, 1-, and 10-cent treatments. Minimum distance (MD) uses treatment means. Nonlinear least squares (NLS) uses individual effort rounded to the nearest 100 points, with scores below 50 assigned to 25, following Appendix A. Power NLS fits log rounded effort; exponential NLS fits effort in levels.

Table 5A: Benchmark parameters and predictions: replication and original paper

	Minimum distance		Nonlinear least squares	
	Replication	Original paper	Replication	Original paper
Power cost				
Curvature γ	33.326	33.137	21.266	20.546
	(21.551)	(10.251)	(5.637)	(5.266)
Cost level k	6.01e−113	2.56e−112	3.45e−72	5.12e−70
	(1.90e−63)	(1.11e−69)	(1.48e−70)	(2.36e−68)
Intrinsic motivation s	6.65e−7	7.13e−7	1.33e−6	3.17e−6
	(1.02e−5)	(9.63e−6)	(3.84e−6)	(8.16e−6)
N	1,664	1,664	1,664	1,664
Implied effort, 4-cent treatment	2,116	2,116	7.586	7.608
Implied effort, low-pay treatment	1,894	1,893	7.413	7.429
Exponential cost				
Curvature γ	0.0159	0.0158	0.0156	0.0156
	(0.0102)	(0.0054)	(0.0039)	(0.0039)
Cost level k	1.05e−16	1.27e−16	1.71e−16	1.71e−16
	(1.69e−11)	(4.87e−12)	(1.41e−15)	(1.41e−15)
Intrinsic motivation s	3.13e−6	3.32e−6	3.72e−6	3.72e−6
	(2.45e−5)	(2.32e−5)	(8.81e−6)	(8.81e−6)
N	1,664	1,664	1,664	1,664
Implied effort, 4-cent treatment	2,117	2,117	2,115	2,115
Implied effort, low-pay treatment	1,884	1,883	1,880	1,885
Discrete pay: normal errors	—	—	1,664	1,881
Discrete pay: empirical errors	—	—	1,651	1,878

Notes: Parentheses contain standard errors. Replication MD standard errors use 1,000 within-treatment bootstrap draws; NLS standard errors are heteroskedasticity-robust (HC1). Intrinsic motivation s is in cents per point. Power NLS predictions are expected log rounded effort; all other predictions are in points. Low pay is 1 cent per 1,000 points. The final two rows give its discrete-pay predictions under exponential NLS. Sources: my Python output and published Table 5, Panel A (p. 1061).

Most estimates and continuous-pay predictions are close to the paper's. Using the paper's rounded treatment means removes the differences in the MD point estimates, and the exponential NLS estimates already match at the reported precision. Power NLS parameter estimates differ, although its continuous-pay predictions remain close to those in the paper. The MD standard errors for γ are roughly twice the published values. Two features of the bootstrap explain the gap. First, the MD bootstrap distributions are sensitive to a few extreme draws, so different draws can produce noticeably different standard errors. Second, the released code is slightly inconsistent with Appendix A. Following the appendix, I resample each treatment at its original size (540, 558 and 566 observations); the code draws 550 from each group, which differs little in size, and also applies additional numerical filters to the estimates.

The main discrepancy in the low-pay predictions stems from differences between the model stated in Appendix A and the authors' released simulation code. A Python translation of the released code predicts average effort of about 1,875 points, close to the published predictions of 1,881 and 1,878. My implementation of the appendix model instead predicts 1,664 points with normal errors and 1,651 with empirical errors—about 12 percent below the published values—even though the underlying exponential NLS parameter estimates are almost identical.

The largest source of this gap is the cost calculation. Following Appendix A, my code uses $c_i(e) = (k_i/\gamma)\exp(\gamma e)$, whereas the released simulation code uses $c_i(e) = k_i \exp(\gamma e)$ without adjusting the definition of k_i, so the factor $1/\gamma$ is missing. Adding it to the released code lowers the empirical-error prediction by about 205 points, most of the 227-point gap. The bonus schedule also differs: at 1,000, 2,000 and 3,000 points, the released code pays 1, 4 and 9 cents, while Appendix A specifies cumulative payments of 1, 2 and 3 cents, as in the experimental design.

The replication supports the paper's finding that monetary incentives increase effort. However, the discrete-pay model in Appendix A predicts a smaller response in the low-pay treatment than the data show.

1.3 Table 5B: social preferences and time preferences

I use the benchmark, charity, gift and delayed-pay treatments to estimate worker preferences. For experts, I recover the parameters implied by their forecasts, holding the benchmark parameters at the worker estimates.

Table 5B: Preference estimates: replication and original paper

	Workers		Experts	
Parameter	Replication	Original paper	Replication	Original paper
Power cost: MD				
Pure altruism α	0.003 (−0.024, 0.034)	0.003 (−0.024, 0.042)	0.067 (0.002, 0.529)	0.068 (0.002, 0.510)
Warm glow a	0.123 (0.001, 0.602)	0.125 (0.003, 0.600)	0.020 (−0.001, 0.703)	0.022 (−8.0e−4, 0.760)
Gift exchange $\Delta s \times 10^6$	3.1 (0, 70.51)	3.26 (−0.07, 80)	12.19 (0.98, 216.12)	13 (2, 200)
Present bias β	1.16 (0.0551, 9.58)	1.17 (0.09, 11)	0.763 (0.306, 1.18)	0.76 (0.27, 1.2)
Weekly discount δ	0.753 (0.317, 1.54)	0.75 (0.33, 1.5)	0.846 (0.613, 0.976)	0.85 (0.62, 1)
Power cost: NLS				
Pure altruism α	0.013 (−0.036, 0.062)	0.006 (−0.024, 0.037)	—	—
Warm glow a	0.259 (−0.236, 0.754)	0.182 (−0.176, 0.540)	—	—
Gift exchange $\Delta s \times 10^6$	12.17 (−40.79, 65.13)	8.16 (−28, 40)	—	—
Present bias β	1.61 (−1.94, 5.15)	1.36 (−1.56, 4.28)	—	—
Weekly discount δ	0.749 (0.244, 1.25)	0.75 (0.261, 1.24)	—	—
Exponential cost: MD				
Pure altruism α	0.003 (−0.024, 0.034)	0.003 (−0.024, 0.041)	0.067 (0.002, 0.538)	0.067 (0.002, 0.530)
Warm glow a	0.140 (0.002, 0.621)	0.143 (0.005, 0.610)	0.028 (−3.0e−4, 0.690)	0.031 (−1.0e−4, 0.710)
Gift exchange $\Delta s \times 10^6$	8.23 (0, 119.9)	8.59 (−0.2, 130)	26.34 (2.95, 299.86)	28 (4.5, 280)
Present bias β	1.15 (0.064, 8.39)	1.15 (0.09, 9.9)	0.77 (0.32, 1.17)	0.76 (0.29, 1.2)
Weekly discount δ	0.765 (0.332, 1.5)	0.76 (0.34, 1.4)	0.853 (0.638, 0.976)	0.85 (0.65, 1)
Exponential cost: NLS				
Pure altruism α	0.004 (−0.018, 0.025)	0.004 (−0.018, 0.025)	0.070 (0.002, 0.539)	0.070 (0.002, 0.530)
Warm glow a	0.143 (−0.132, 0.417)	0.143 (−0.132, 0.417)	0.034 (4.2e−5, 0.724)	0.035 (4.5e−5, 0.740)
Gift exchange $\Delta s \times 10^6$	9.4 (−27.39, 46.18)	9.40 (−28, 48)	29.59 (3.3, 327.85)	30 (5, 300)
Present bias β	1.24 (−1.30, 3.77)	1.24 (−1.3, 3.77)	0.806 (0.339, 1.22)	0.79 (0.3, 1.2)
Weekly discount δ	0.754 (0.283, 1.22)	0.75 (0.283, 1.22)	0.855 (0.642, 0.977)	0.86 (0.65, 1)

Notes: Intervals below worker estimates are 95% confidence intervals; those below expert medians show the 25th and 75th percentiles. Worker NLS intervals use HC1 standard errors and a normal approximation. Published NLS gift coefficients and confidence intervals are multiplied by 0.4 to express them as total shifts. All gift-shift values are then multiplied by 10^6.

Most of Panel B agrees closely with the paper. Small differences in the MD point estimates reflect rounding of treatment means; differences in their confidence intervals reflect the bootstrap procedures discussed for Panel A. Expert summaries differ slightly because of the inclusion rules: I retain all 208 experts for social preferences but exclude nine whose delay forecasts imply no positive solution for β and δ from the time-preference rows.

Two NLS results needed further checks. The exponential NLS gift shift matches once the published coefficient is multiplied by 0.4, expressing it as a total shift, as in the MD columns. The larger difference is in power NLS. Starting from the authors' saved coefficients, re-estimating the same specification lowers the residual sum of squares from 1,542.14 to 1,539.45 and recovers my estimates. This points to a numerical optimization issue in the archived results.

The substantive conclusions are unchanged. Worker estimates suggest little pure altruism and positive warm glow, while experts expect more altruism. For workers, $\hat{\beta} > 1$, but the confidence intervals include 1, providing no clear evidence of present bias. Experts' median implied β is about 0.8, below 1, indicating that they expect present bias.

2. Extension: predicting effort at the highest piece rate

The paper estimates its MD models on the 0-, 1- and 10-cent treatments and predicts effort at 4 cents, which tests interpolation within the range of incentives used in estimation. Here I reverse the roles of the two highest pay rates. Using the same MD estimator and the same two cost functions, I estimate the models on the 0-, 1- and 4-cent treatments and predict mean effort at 10 cents, outside the estimation range. Table E1 reports the results.

Table E1: Extrapolation to the held-out 10-cent treatment

Cost function	Predicted mean	Observed mean	Error	Error (%)	95% CI for error
Power	2,202.38	2,174.62	27.76	1.28	[−74.74, 136.03]
Exponential	2,199.58	2,174.62	24.96	1.15	[−75.17, 129.97]

Notes: Estimation uses 1,660 workers; the holdout contains 566. Error is predicted minus observed mean effort; percentage error uses the observed mean as the denominator. The 95% percentile intervals use 998 valid draws out of 1,000, resampling within all four treatments and re-estimating the model.

Both models predict mean effort at 10 cents with errors below 1.3%, compared with about 0.7–0.8% in the paper's interpolation exercise. Both confidence intervals for the prediction error include zero. These results support the robustness of mean-effort predictions to this alternative treatment split: estimation using only lower-pay groups still predicts the highest-pay group closely, under both cost functions.

Reference. DellaVigna, Stefano, and Devin Pope (2018). "What Motivates Effort? Evidence and Expert Forecasts." *Review of Economic Studies* 85(2): 1029–1069.